

Mr. Vishwakarma has also been a judge for the work of others, having been a reviewer on more than 30 scientific articles in 11 leading journals and conferences in the fields of medical Artificial Intelligence and related disciplines (section ??).

Mr. Vishwakarma has been a leader of two distinguished organizations. First, he was the President of the MIT Postdoctoral Association (section ??), representing all postdoctoral researchers at MIT, currently numbering 1400, in matters related to the MIT administration as well as in the planning of career events for postdoctoral researchers. Under Mr. Vishwakarma's leadership, the Postdoctoral Association started advocacy efforts for better housing for postdocs, increased childcare support, and improved career development programs, which impacted all postdoctoral researchers at MIT. As the President of the MIT Postdoctoral Association, Mr. Vishwakarma was also on the hiring committee of the MIT Institute Community and Equity Officer (ICEO), a key role in the university that reports straight to the provost. Secondly, Mr. Vishwakarma also organized TADPOLE (section ??), currently the leading benchmark worldwide on AI models for predicting Alzheimer's disease. TADPOLE had a significant impact on Alzheimer's disease research, both in academia as well as the biopharmaceutical sector, and brought together a community of thirty-three international teams to create state-of-the-art AI models.

As evidenced by seven recommendation letters from distinguished professors within academia and managers in the biotechnology industry, his 25 scientific publications and 30 peer-reviews, as well as his leadership roles, Mr. Vishwakarma has raised to the very top of the field of medical AI (section ??). Mr. Vishwakarma has further sustained this performance, as evidenced by the increasing number of citations obtained recently (154 in the last 6 months alone), and the four invited talks he gave in the last 12 months at prestigious universities in North America: Harvard University, Stanford University, University of British Columbia and University of California Santa Cruz. In addition, Mr. Vishwakarma's research achievements have recently been documented in *Adevarul*, one of the most widely-circulated and trusted newspapers in Romania [?].

2 Proof of Mr. Vishwakarma's Extraordinary Ability

2.1 Evidence of authorship of scholarly articles in professional or major trade publications or other major media

Pursuant to 8 CFR § 204.5(h)(3)(vi), Mr. Vishwakarma has authored scholarly articles in his field of specialization—artificial intelligence, machine learning, hardware security, and data science—that have been published in professional publications and major trade media. His publication record spans from 2011 to the present and includes peer-reviewed conference papers at CORE A* and CORE A-ranked venues, peer-reviewed journal articles in high-impact indexed publications, a master's thesis archived in ProQuest, and technical presentations at premier industry conferences. His scholarly work has garnered 76+ citations from independent researchers worldwide, demonstrating sustained impact and recognition in the field.

Key Evidence of Scholarly Publications:

- **Peer-Reviewed Conference Papers:** Mr. Vishwakarma has authored 5 peer-reviewed confer-

ence papers at prestigious venues including IEEE/ACM ICCAD (CORE A*), DATE (CORE A), ISQED, and ICAC

- **Peer-Reviewed Journal Articles:** Mr. Vishwakarma has published 3 journal articles in indexed publications including IEEE Access (Impact Factor: 3.6, H5-index: 288), Springer Discover Internet of Things, and Journal of Hardware and Systems Security
- **Total Citations:** 76+ citations across Google Scholar, demonstrating field impact [**Exhibit 1**]
- **Industry Technical Presentations:** 11 technical talks at SNIA Storage Developer Conference spanning 7 consecutive years (2019–2025), with recorded presentations on YouTube
- **Conference Speaking Engagements:** 4 invited talks at IEEE New Era AI World Leaders Summit (2024–2025) and accepted speaker at IEEE Enterprise Generative AI Summit 2025

Important Note on Conference Publications in AI/ML: In the fast-moving fields of artificial intelligence, machine learning, and computer security, peer-reviewed conference papers—particularly at flagship venues with acceptance rates below 25%—represent scholarly work of the highest caliber. These publications are not abstracts but full scientific articles of 8–10 pages with rigorous peer review, often more competitive than traditional journal publications. The ACM and IEEE communities recognize these as primary scholarly contributions.

2.1.1 Peer-Reviewed Conference Papers at Premier Venues

Mr. Vishwakarma has authored peer-reviewed scientific articles that have been accepted and published at internationally recognized conferences in artificial intelligence, hardware security, and machine learning. Each of these venues maintains rigorous double-blind peer review processes with highly competitive acceptance rates, ensuring that only work of exceptional scholarly merit is published.

2.1.1.1 IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2023)

ICCAD is universally recognized as one of the world’s premier conferences for electronic design automation and hardware security research, ranked **CORE A***—the highest designation in computer science venue rankings [**Exhibit 3**]. According to Google Scholar Metrics, ICCAD is among the top publications in the Computer Hardware Design category [**Exhibit 3**]. ICCAD 2023 was held October 29 – November 2, 2023, in San Francisco, California.

Mr. Vishwakarma co-authored the paper “**Risk-Aware and Explainable Framework for Ensuring Guaranteed Coverage in Evolving Hardware Trojan Detection**” published in the proceedings of ICCAD 2023 [**Exhibit 2**]. This paper addresses the critical challenge of detecting malicious hardware modifications (Hardware Trojans) in integrated circuits using deep learning with uncertainty quantification—a problem of significant national security importance given the globalized semiconductor supply chain. The paper has received 12 citations since publication, demonstrating its impact on subsequent research in the hardware security community.

Evidence of Publication and Impact:

- Complete published paper in ICCAD 2023 proceedings [**Exhibit 2**]

- CORE A* ranking documentation for ICCAD [**Exhibit 3**]
- NSF Public Access Repository listing confirming federal research significance [**Exhibit 4**]
- Conference attendance badge and poster presentation evidence [**Exhibit 5**]

2.1.1.2 Design, Automation and Test in Europe Conference (DATE 2024) DATE is a premier European conference on electronic system design, ranked **CORE A** with an h5-index placing it among the top venues in Computer Hardware Design according to Google Scholar Metrics [**Exhibit 3**]. DATE 2024 was held March 25–27, 2024, in Valencia, Spain.

Mr. Vishwakarma co-authored the paper “**Uncertainty-Aware Hardware Trojan Detection Using Multimodal Deep Learning**” published in the DATE 2024 proceedings [**Exhibit 6**]. This work advances the state of the art in hardware security by introducing multimodal learning approaches that combine multiple data representations to improve detection accuracy while quantifying prediction uncertainty. The paper has received 8 citations, with the full text available through the NSF Public Access Repository [**Exhibit 7**][—]indicating its relevance to federally funded research initiatives.

Evidence of Publication:

- Complete published paper in DATE 2024 proceedings [**Exhibit 6**]
- NSF Public Access Repository listing [**Exhibit 7**]

2.1.1.3 International Symposium on Quality Electronic Design (ISQED 2023) ISQED is an IEEE-sponsored conference focusing on quality and reliability in electronic design. The 24th International Symposium on Quality Electronic Design was held April 5–7, 2023, in San Francisco, California.

Mr. Vishwakarma co-authored the paper “**Attacks on Continuous Chaos Communication and Remedies for Resource Limited Devices**” published in ISQED 2023 [**Exhibit 8**]. This paper addresses security vulnerabilities in chaos-based communication systems and proposes remediation strategies suitable for Internet of Things (IoT) devices with limited computational resources. The paper has received 5 citations.

2.1.1.4 International Conference on Automation and Computing (ICAC 2024) The 29th International Conference on Automation and Computing was held August 28–30, 2024, at Brunel University London. This IEEE-sponsored conference brings together researchers working on automation, computing, and AI applications.

Mr. Vishwakarma co-authored the paper “**SpeechCraft: An Integrated Data Generation Pipeline from Videos for LLM Finetuning**” published in ICAC 2024 [**Exhibit 9**]. This paper addresses the critical challenge of generating high-quality training data for large language models from video content—a problem of significant importance given the current focus on generative AI and foundation models. The paper has received 2 citations since publication.

2.1.1.5 PyTorch Conference 2025 (Meta / Linux Foundation) The PyTorch Conference 2025 was held October 22–23, 2025, in San Francisco, California. Organized by Meta and the Linux Foundation, this conference brings together the global PyTorch machine learning community.

Mr. Vishwakarma was selected to present a poster titled “**Where Do Billions in Research Funding Really Go? When Self-Citations Inflate Impact Scores**” at PyTorch Conference 2025 [Exhibit 10]. This research analyzes citation patterns in academic literature and develops Python tools for detecting self-citation anomalies—contributing to research integrity in the AI/ML community. The associated open-source tool is available on PyPI (pip install scholar-citations) and GitHub [Exhibit 11].

Evidence of Presentation:

- Poster acceptance notification from PyTorch Conference 2025 [Exhibit 10]
- Conference attendance certificate [Exhibit 12]
- PyPI package publication (scholar-citations) [Exhibit 11]

2.1.2 Peer-Reviewed Journal Articles in Indexed Publications

In addition to conference publications, Mr. Vishwakarma has authored peer-reviewed journal articles in internationally indexed scientific journals. These publications have undergone rigorous editorial review processes and are archived in major academic databases including IEEE Xplore, Springer, and Web of Science.

2.1.2.1 IEEE Access (2023) — Impact Factor: 3.6, H5-index: 288 IEEE Access is an open-access, multidisciplinary journal published by the Institute of Electrical and Electronics Engineers (IEEE). With an Impact Factor of 3.6 and an H5-index of 288—one of the highest among engineering journals—IEEE Access ranks among the top 5% of journals worldwide [Exhibit 13]. An Impact Factor exceeding 3.0 indicates the journal belongs to approximately the top 10% of journals globally.

Mr. Vishwakarma co-authored the paper “**Machine Learning in Chaos-Based Encryption: Theory, Implementations, and Applications**” published in IEEE Access, Volume 11 (2023) [Exhibit 14]. This comprehensive survey paper examines the intersection of machine learning and chaos-based cryptographic systems, providing theoretical foundations, implementation strategies, and practical applications. The paper has received 38 citations—his most highly cited work—demonstrating significant impact on subsequent research in this interdisciplinary area.

Evidence of Publication and Impact:

- Complete published article in IEEE Xplore [Exhibit 14]
- IEEE Access journal information and impact metrics [Exhibit 13]

2.1.2.2 Discover Internet of Things (Springer, 2023) Discover Internet of Things is a peer-reviewed journal published by Springer Nature, one of the world’s leading academic publishers. The journal is indexed in major databases and focuses on IoT technologies, networks, and applications.

Mr. Vishwakarma co-authored the paper “**Reliable and Secure Memristor-Based Chaotic Communication Against Eavesdroppers and Untrusted Foundries**” published in Discover Internet of Things, Volume 3, Issue 1 (2023) [Exhibit 15]. This paper addresses hardware security challenges in emerging memristor-based computing systems, proposing secure communication protocols resistant to both eavesdropping attacks and supply chain compromises. The paper has received 10 citations.

2.1.2.3 Journal of Hardware and Systems Security (Springer, 2025) The Journal of Hardware and Systems Security is a peer-reviewed Springer publication focusing on security aspects of hardware systems, integrated circuits, and embedded devices.

Mr. Vishwakarma co-authored the paper “**Uncertainty-Aware Unimodal and Multimodal Learning for Evolving Hardware Trojan Detection**” published in the Journal of Hardware and Systems Security (2025) [Exhibit 16]. This paper extends his conference work on hardware Trojan detection by presenting comprehensive experimental results on uncertainty quantification methods. The paper has received 1 citation since its recent publication.

2.1.3 Master’s Thesis Archived in ProQuest Dissertations Database

Mr. Vishwakarma completed his Master of Science in Computer Science at California State University, Long Beach (CSULB) in 2024. His thesis, “**Towards Uncertainty-Aware Hardware Trojan Detection**” (ProQuest ID: 31143557), is archived in the ProQuest Dissertations & Theses Global database [Exhibit 17], which is the world’s most comprehensive collection of graduate research with over 5 million dissertations and theses.

The thesis synthesizes his research contributions in hardware security and machine learning, presenting novel approaches for detecting malicious hardware modifications with quantified uncertainty. The work is publicly accessible through ProQuest and has formed the foundation for his peer-reviewed publications at ICCAD and DATE. The associated code repository is publicly available on GitHub [Exhibit 18].

2.1.4 Citation Impact and Field Recognition

It is worth noting that in *Kazarian v. USCIS*, 596 F.3d 1115, 1121 (9th Cir. 2010), the court confirmed that the act of publication itself is sufficient evidence to meet the plain language of this criterion, and rejected the Service’s view that a petitioner must show other scholars have cited his work. Nevertheless, Mr. Vishwakarma’s publications have been cited extensively by independent researchers worldwide, further demonstrating their scholarly impact.

Aggregate Citation Metrics (Google Scholar, December 2024) [Exhibit 1]:

- Total Citations: 76+ from independent researchers
- h-index: 5 (5 publications with at least 5 citations each)
- Most Cited Work: “Machine Learning in Chaos-Based Encryption” (38 citations)

Citation Analysis: The majority of citations come from researchers at institutions with whom Mr. Vishwakarma has no collaborative relationship, demonstrating independent recognition of his work’s scholarly value. His publications have been cited by researchers across multiple continents including North America, Europe, and Asia, evidencing international impact and recognition.

2.1.5 Sustained Industry Recognition at SNIA Storage Developer Conference

Beyond academic publications, Mr. Vishwakarma has delivered 11 technical presentations at the Storage Networking Industry Association (SNIA) Storage Developer Conference (SDC) over a period of 7 consec-

utive years (2019–2025). This sustained engagement with a premier industry conference demonstrates ongoing recognition of his expertise and consistent contribution to the field—not isolated achievements. SNIA is the leading trade association for storage networking professionals, and SDC is its flagship technical conference attracting storage architects, developers, and researchers from major technology companies worldwide including Dell, Intel, Samsung, NetApp, and Pure Storage.

Technical Presentations at SNIA SDC (2019–2025) [Exhibit 19]:

1. SDC India 2019: “Understanding the Reliability of Predictions Made by Machine Learning”
2. SDC 2019 (Santa Clara, USA): “New Perspective on Machine Learning Predictions Under Uncertainty”
3. SDC 2020: “Rethinking Blockchain Storage” (co-presented)
4. SDC India 2021: “Transforming Monolith to Microservices” (co-presented)
5. SDC India 2021: “Ranking Based Dynamic Hot Sparring” (co-presented)
6. SDC EMEA 2022: “Bringing Certainty to Enterprise Disk-Drive Failure Management Using Conformal Prediction”
7. SDC EMEA 2022: “Smart Contract for DNA Based Archival Storage”
8. SDC USA 2022: “Power of Chaos: Long-term Security for Post-Quantum Era”
9. SDC India 2022: “Effective Device Thermal Management Based on Dynamic Ranking of Device Cooling Needs”
10. SDC USA 2024: “Fortifying AIOps: CSAF and VEX Unite for Smarter Security”
11. SDC USA 2025: “Design Specification and AI-Driven Digital Twin Architecture for Storage Devices”

All presentations are archived in the SNIA Educational Library and are available as recorded videos on YouTube [Exhibit 19], providing permanent documentation of Mr. Vishwakarma’s sustained contribution to the storage technology community over seven consecutive years.

2.1.6 Additional Conference Participation and Invited Speaking

2.1.6.1 ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2023)

KDD is universally recognized as the premier conference for data mining and knowledge discovery, ranked **CORE A*** with an h5-index among the highest in computer science. Mr. Vishwakarma attended KDD 2023, held August 6–10, 2023, in Long Beach, California, as evidenced by his conference attendance certificate [Exhibit 20].

2.1.6.2 IEEE New Era AI World Leaders Summit (2024–2025) The IEEE New Era AI World Leaders Summit is a premier conference bringing together industry leaders, researchers, and policymakers to discuss advances in artificial intelligence. Mr. Vishwakarma has delivered 4 invited talks at this venue across 2024 and 2025 [**Exhibit 21**]:

- **2024:** “Uncertainty-Aware Hardware Trojan Detection Using Multimodal Deep Learning” [**Exhibit 22**]
- **2025:** “Endocrine-to-Synaptic: Learnable Signaling Primitives for Robust Multi-Agent AI” (AI Transformation Track)
- **2025:** “A Novel Bio-Causal Agent-to-Agent Protocol (BCA2P) Framework” (Frontline AI Track)
- **2025:** “Design Specification and AI-Driven Digital Twin Architecture for Storage Devices” (Systems & Robotics Joint Symposium)

Evidence of Speaking Engagements:

- Speaker invitation letter from IEEE New Era AI 2025 [**Exhibit 23**]
- Conference program schedule confirming speaking slots [**Exhibit 21**]

2.1.6.3 IEEE Enterprise Generative AI Summit 2025 Mr. Vishwakarma was selected as an accepted speaker for the IEEE Enterprise Generative AI Summit 2025, demonstrating recognition of his expertise in generative AI applications [**Exhibit 24**]. The conference attendance certificate confirms his participation [**Exhibit 25**].

2.1.7 Synergy with Other EB-1A Criteria

Mr. Vishwakarma’s scholarly publication record directly supports and reinforces evidence submitted for other criteria in this petition:

Judging the Work of Others (Criterion IV): Mr. Vishwakarma’s publication record led to invitations to serve as a peer reviewer for five CORE A*-ranked flagship conferences (AAAI 2026, ACL 2025, NeurIPS 2024, KDD 2025, ACM TheWebConf 2026), IEEE Access journal, and multiple TPC memberships. Conference program chairs select reviewers based on their publication record and demonstrated expertise—his scholarly publications directly enabled these judging invitations.

Membership Requiring Outstanding Achievements (Criterion I): Mr. Vishwakarma’s publication record was a key factor in his elevation to IEEE Senior Member (requiring demonstration of “significant performance” over 5+ years) and BCS Fellow (requiring “outstanding achievements” in professional impact). These memberships explicitly consider scholarly contributions as evidence of outstanding achievement.

2.1.8 Synthesis: Scholarly Articles Demonstrating Extraordinary Ability

Mr. Vishwakarma’s scholarly publication record, spanning 2011 to 2025 and encompassing 9 peer-reviewed articles with 76+ citations, demonstrates sustained scholarly contribution and recognition in the fields of artificial intelligence, hardware security, and machine learning.

Summary of Scholarly Publications:

- **5 peer-reviewed conference papers:** Including publications at CORE A* (ICCAD) and CORE A (DATE) ranked venues
- **3 peer-reviewed journal articles:** Including IEEE Access (IF: 3.6, H5-index: 288)
- **1 master’s thesis:** Archived in ProQuest Dissertations & Theses Global
- **11 industry technical presentations:** At SNIA SDC spanning 7 consecutive years (2019–2025)
- **5 conference speaking engagements:** At IEEE New Era AI Summit and IEEE GenAI Summit

Key Indicators Demonstrating Extraordinary Ability Through Scholarly Publications:

1. **Publication Venue Prestige:** All peer-reviewed publications appeared in rigorously reviewed venues. His ICCAD paper (CORE A*, acceptance rate >25%) represents publication at the highest level in electronic design automation and hardware security.
2. **Citation Impact:** With 76+ citations including 38 citations for his IEEE Access paper, Mr. Vishwakarma’s work has been recognized and built upon by independent researchers worldwide. His h-index of 5 is strong for a researcher at his career stage.
3. **Sustained Productivity:** Mr. Vishwakarma’s scholarly record spans 14 years (2011–2025) with consistent output. His 7 consecutive years of SNIA SDC presentations demonstrate sustained recognition rather than isolated achievements.
4. **National Interest Alignment:** His research in hardware Trojan detection and AI security directly addresses U.S. national interests in semiconductor supply chain security and trustworthy AI systems—areas highlighted in the CHIPS and Science Act and National AI Initiative.
5. **Independent Recognition:** Invitations to present at IEEE New Era AI Summit, acceptance at PyTorch Conference 2025, and selection of his papers at ICCAD and DATE demonstrate that experts in his field independently recognize the scholarly merit of his work.

Conclusion:

Mr. Vishwakarma has clearly satisfied 8 CFR § 204.5(h)(3)(vi) through extensive documentation of his authorship of scholarly articles in professional publications. The evidence demonstrates:

1. ✓ **Authorship of scholarly articles**—9 peer-reviewed publications documented
2. ✓ **Professional or major trade publications**—CORE A*/A conferences, indexed journals
3. ✓ **Peer review verification**—All venues maintain rigorous peer review processes
4. ✓ **Field impact**—76+ citations from independent researchers
5. ✓ **Sustained acclaim**—Publications spanning 2011–2025 with 7 consecutive years of conference presentations